Multiple comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Statistical tests and assumptions. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of multiple comparisons. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **multiple comparisons**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **multiple comparisons**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Statistical tests and assumptions. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **multiple comparisons** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Multiple comparisons case. Separate observations, given facts and assumptions before choosing an explanation.

Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Multiple comparisons. Connect each labelled input, deciding step and result.

For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the **family error** by a bound without requiring independent tests.

Multiple comparisons — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.	For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.	Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Multiple comparisons. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: The declared family has ten tests instead of six; keep the listed values unchanged.

Predicted result

Prediction: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

Why the result follows

Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Multiple comparisons to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Multiple comparisons	Testing several hypotheses in one declared family
Family error	The chance of at least one false rejection under relevant nulls
Bonferroni rule	Dividing a family threshold by the number of tests

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Multiple comparisons, predict a step, then compare with its explanation.

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.

Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Worked explanation · Multiple comparisons

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Multiple comparisons

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def bonferroni(p_values, alpha=0.05):
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in p_values)):
        raise ValueError('test settings')
    threshold = alpha / len(p_values)
    return (threshold, [p <= threshold for p in p_values])

print(bonferroni([.009, .012, .3, .4, .8]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Multiple comparisons

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

The facts we will use

Supplied facts

Familywise alpha=0.05. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

AI ACADEMY / AI-7/15

Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/15

Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count the planned tests

There are six tests in the declared family.



2. Compute corrected threshold

0.05/6 is about 0.00833 for each test.



3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count the planned tests	There are six tests in the declared family.
2. Compute corrected threshold	0.05/6 is about 0.00833 for each test.
3. Apply all six decisions	Only 0.004 is below 0.00833; the other five do not reject under this rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

What this example does not establish

Keep the stated limit

No effect size or real data is supplied. Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

Keep the stated limit

Read the labelled picture: Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

Keep the stated limit

Change one thing: The declared family has ten tests instead of six; keep the listed values unchanged. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Multiple comparisons claim and explain a repair using the supplied evidence.

Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Multiple comparisons with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Multiple comparisons**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Multiple comparisons**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Multiple comparisons. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

```
def bonferroni(p_values, alpha=0.05):
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in p_values)):
        raise ValueError('test settings')
    threshold = alpha / len(p_values)
    return (threshold, [p <= threshold for p in p_values])
```

```
print(bonferroni([.009, .012, .3, .4, .8]))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Multiple comparisons. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **multiple comparisons evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **multiple comparisons**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	

Project evidence	Record
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Multiple comparisons case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Multiple comparisons — evidence trace

INPUT AND SCOPE

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.

CHECK OR CALCULATION

For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.

CONCLUSION AND LIMIT

Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Multiple comparisons, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

By the end, I can	How I will show it
Explain the mechanism and terminology of multiple comparisons	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Multiple comparisons without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **multiple comparisons**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only 0.004 is below 0.00833; the other five do not reject under this rule.

1. Apply all six decisions

2. Compute corrected threshold

3. Count the planned tests

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply all six decisions Only 0.004 is below 0.00833; the other five do not reject under this rule.

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The declared family has ten tests instead of six; keep the listed values unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Repair a missing building block

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

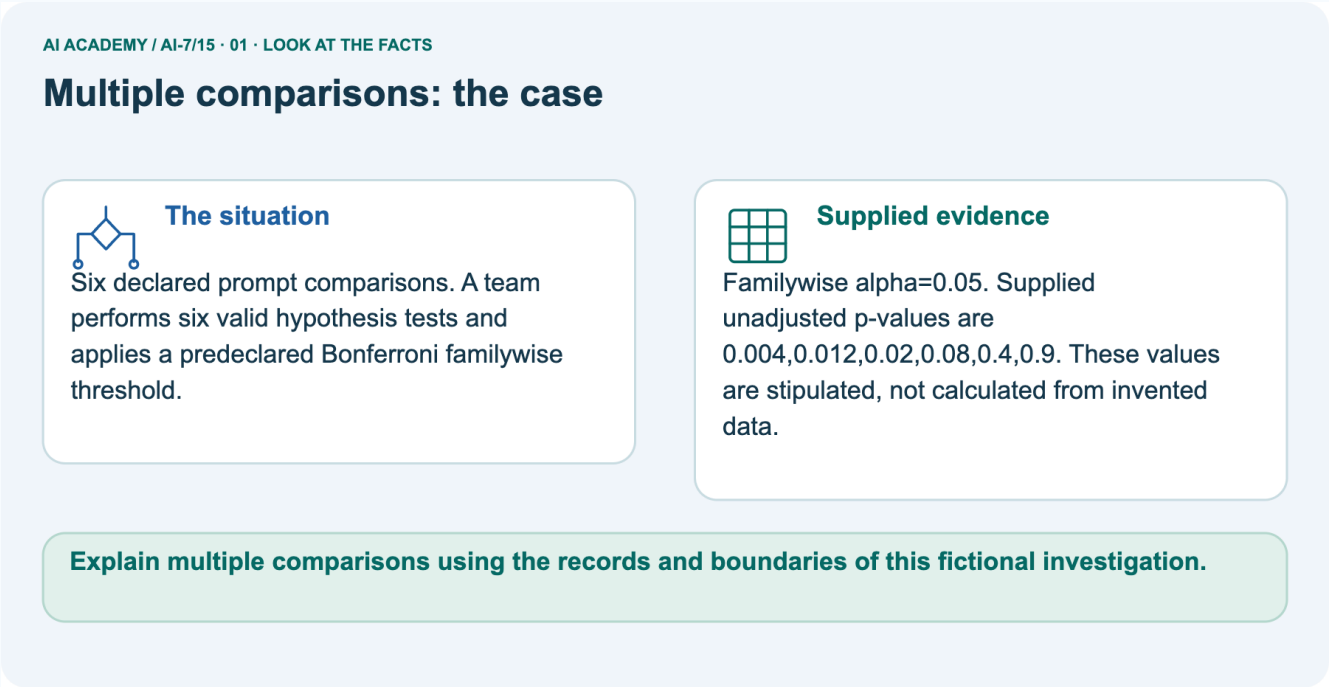
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Multiple comparisons graphic using its labels, deciding evidence and relevant limitation.

01. Multiple comparisons: the case



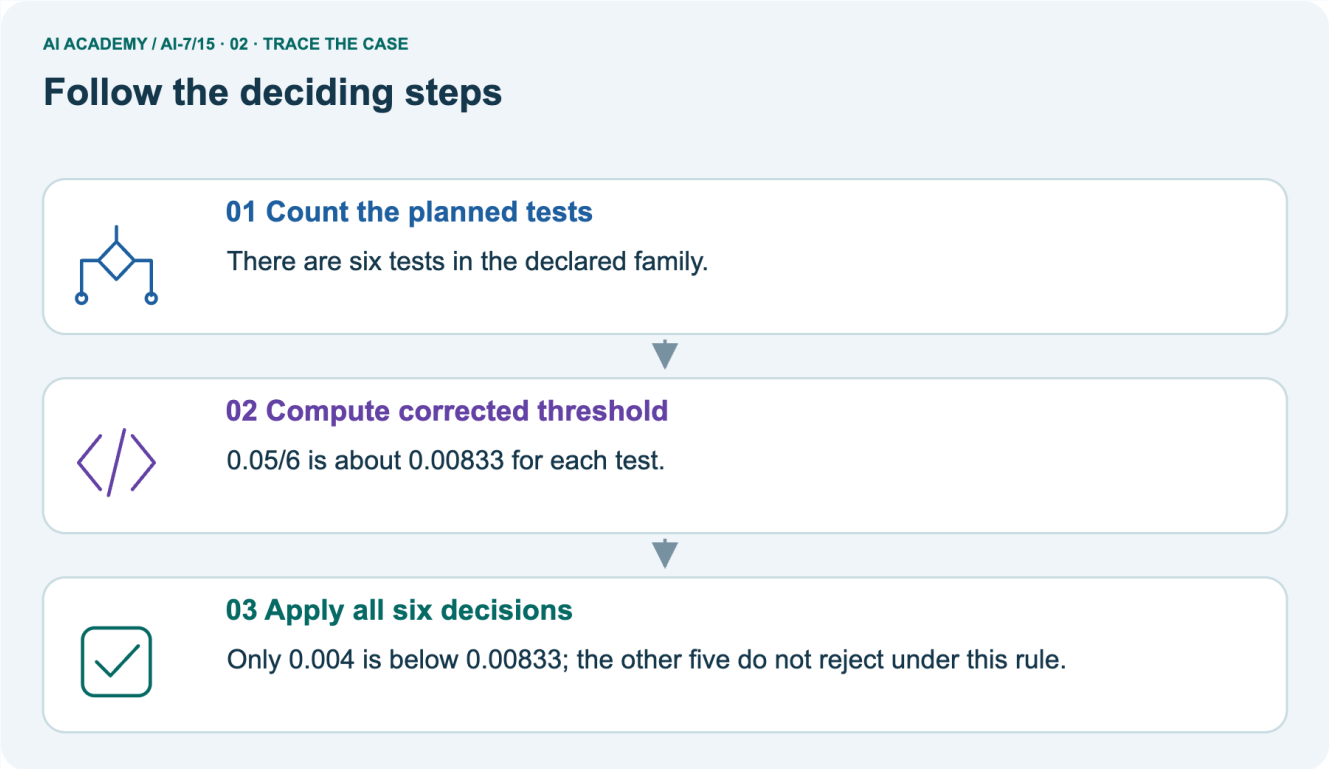
Follow the labelled picture with the matching explanation. Open the vector diagram

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold. Supplied facts: Familywise alpha=0.05.

Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

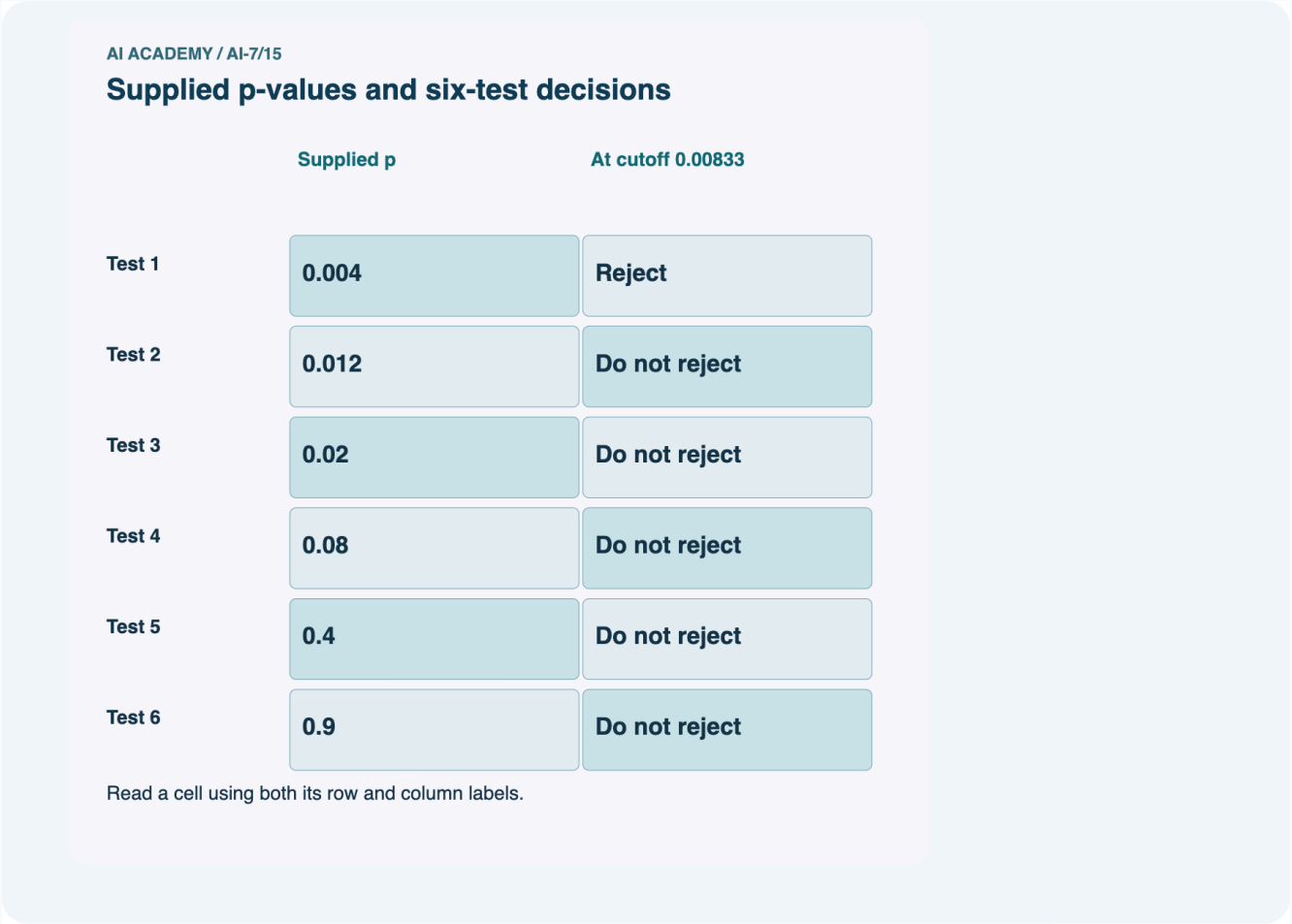


Follow the labelled picture with the matching explanation. Open the vector diagram

Count the planned tests: There are six tests in the declared family. → Compute corrected threshold: 0.05/6 is about 0.00833 for each test. → Apply all six decisions: Only 0.004 is below 0.00833; the other five do not reject under this rule.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Supplied p-values and six-test decisions



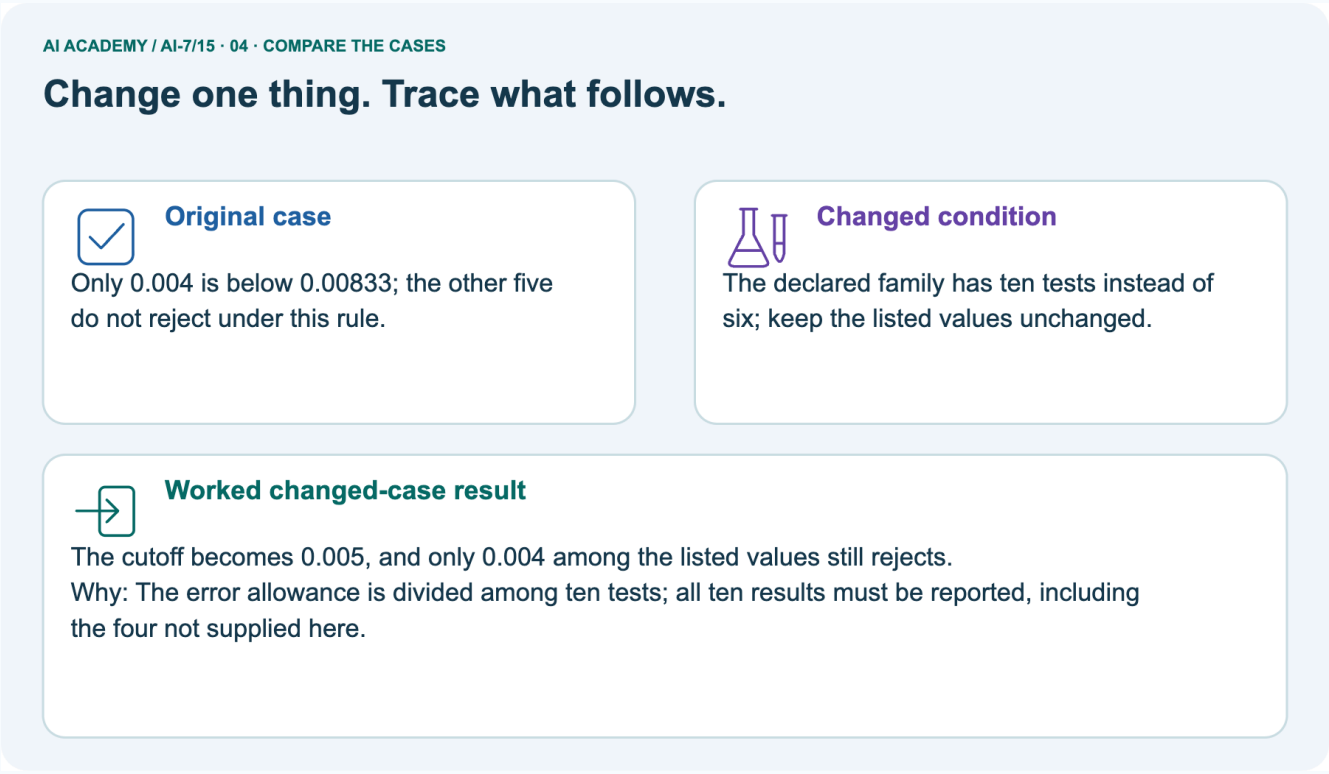
Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied p-values and six-test decisions. Test 1: Supplied p = 0.004, At cutoff 0.00833 = Reject. Test 2: Supplied p = 0.012, At cutoff 0.00833 = Do not reject. Test 3: Supplied p = 0.02, At cutoff 0.00833 = Do not reject.

Test 4: Supplied p = 0.08, At cutoff 0.00833 = Do not reject. Test 5: Supplied p = 0.4, At cutoff 0.00833 = Do not reject. Test 6: Supplied p = 0.9, At cutoff 0.00833 = Do not reject.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



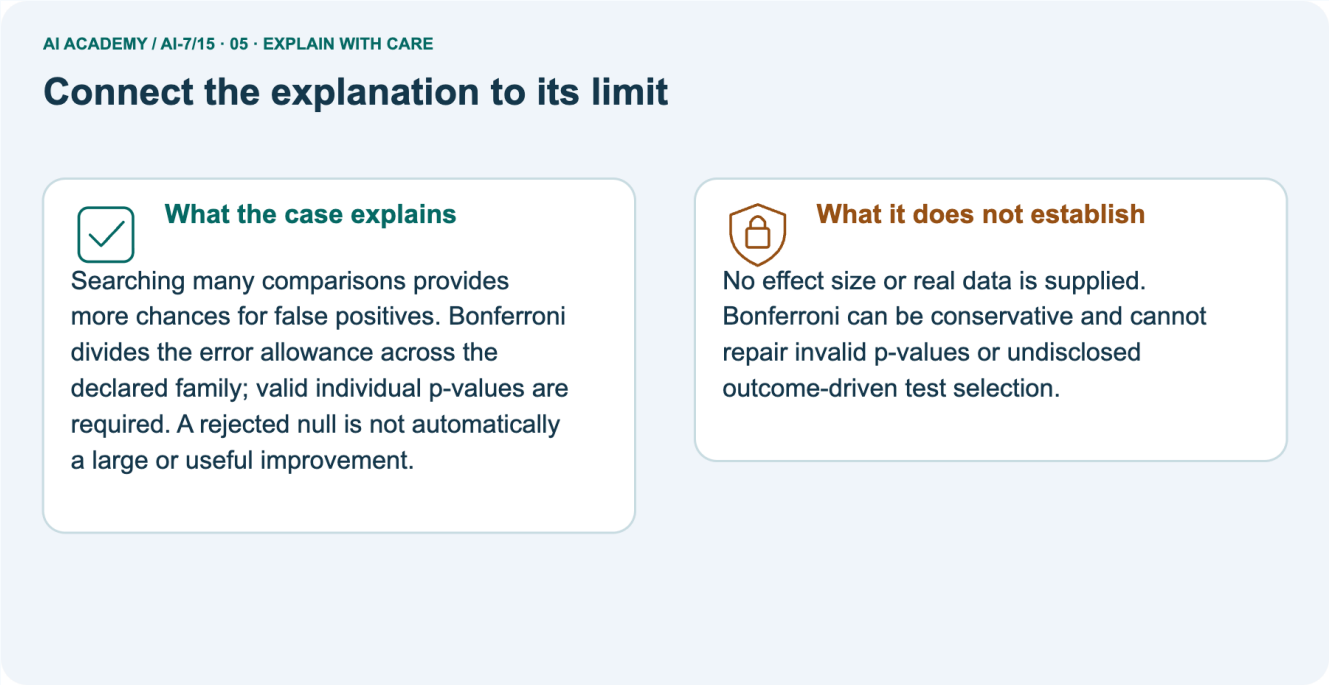
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.004 is below 0.00833; the other five do not reject under this rule. Changed condition: The declared family has ten tests instead of six; keep the listed values unchanged.

Expected result: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

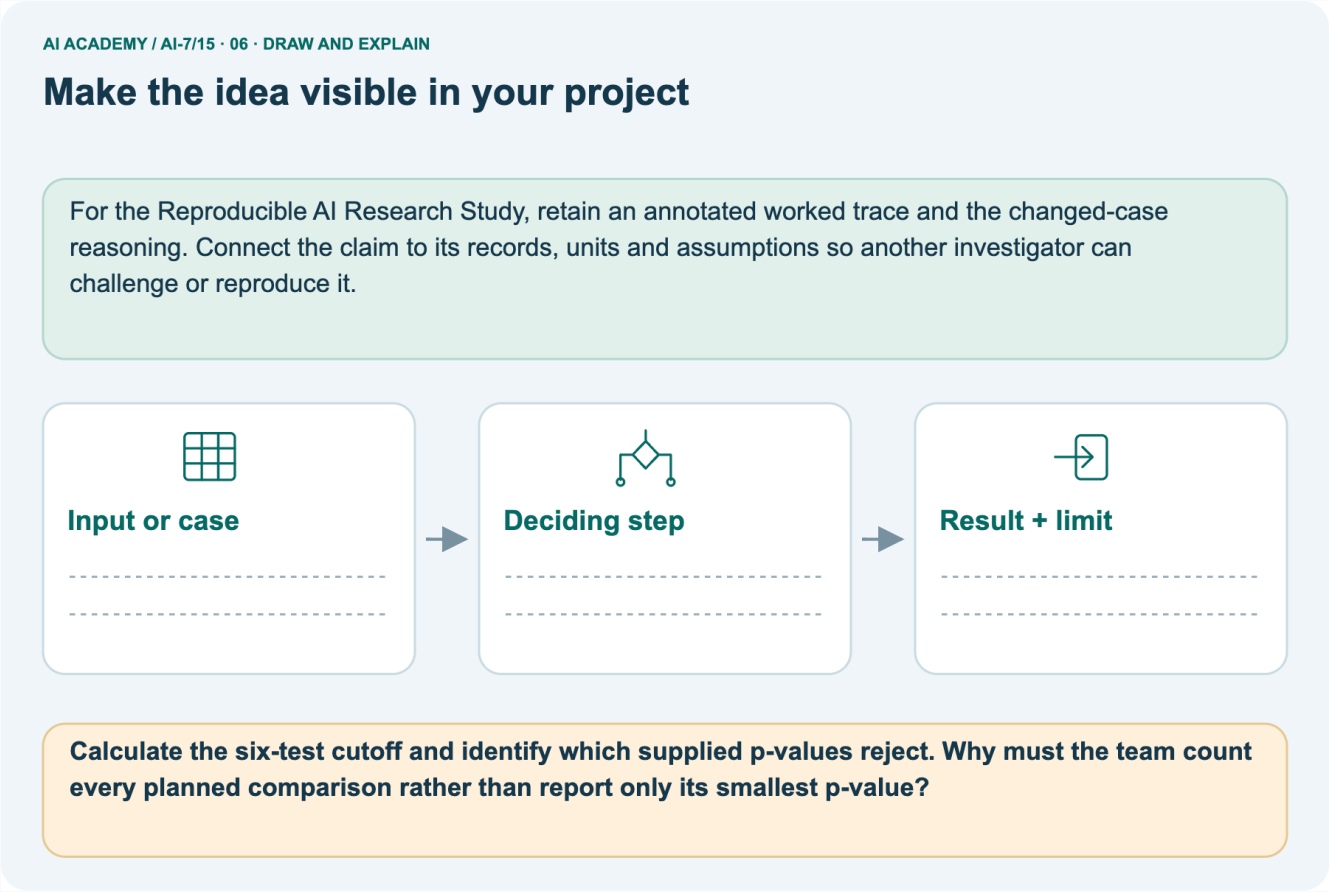
Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement. Limit: No effect size or real data is supplied.

Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.